

Vansh Singh

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SUMMARY

Computer Science graduate from the University of Alberta with hands-on experience building data-cleaning and standardization pipelines, testing full-stack applications, and integrating AI/ML tools using **Python, Pandas, NumPy, JavaScript, and FastAPI**. Experienced aligning inconsistent, semi-structured data into normalized schemas, building queryable search interfaces, and validating software through unit and integration testing on Git-based, Unix/Linux development workflows. Strong academic foundation in Software Engineering, Algorithms & Data Structures, Machine Learning, AI, and Web Development.

EDUCATION

University of Alberta <i>Bachelor of Science in Computing Science, Minor in Film Studies</i>	Edmonton, AB Aug. 2021 – Apr. 2026
• Relevant Coursework: Artificial Intelligence, Algorithms I, Search and Knowledge Simulation, Practical Programming Methodology, File and Database Management, Operating Systems, Parallel and Distributed Systems, Computer Organization and Architecture, Formal Systems and Logic, and Non-Procedural Programming Languages. • Awards: University of Alberta International Student Scholarship; Runner-Up, Calgary Youth Hackathon for an accessibility-focused mental-health platform.	

EXPERIENCE

Shop Squad <i>Software Developer Intern</i>	Remote / Singapore May 2023 – Aug. 2023
• Developed production mobile and backend features using React Native, JavaScript, Node.js, and REST APIs within Agile development workflows; integrated authentication, payment, and API-backed application services. • Debugged frontend, backend, and API-integration issues and contributed to software testing, feature validation, documentation, code reviews, Git-based collaboration, and iterative production improvements.	
University of Alberta Library <i>Library Specialized Technical Support</i>	Edmonton, AB Feb. 2024 – Apr. 2024
• Resolved more than 100 software, hardware, networking, operating-system, and multimedia technology issues across Windows and Linux environments using structured troubleshooting, testing, debugging, and technical documentation. • Maintained public workstations, VR systems, printers, scanners, projectors, and digital learning technologies through system verification, issue tracking, configuration, and preventative maintenance while supporting students, researchers, staff, and other users.	
University of Alberta Film Society <i>Treasurer</i>	Edmonton, AB Sep. 2024 – Apr. 2025
• Managed budgeting, reimbursements, financial records, planning documents, and stakeholder communication while collaborating with executive members to deliver screenings, discussions, and campus events. • Coordinated outreach, event promotion, volunteer activities, and organizational workflows across a multidisciplinary student team.	

PROJECTS

FrameVault Python, FastAPI, React, BeautifulSoup, SQLite, REST APIs, Axios
• Designed a data-cleaning and standardization pipeline that ingests inconsistently structured source data, then normalizes, validates, deduplicates, and indexes it into a consistent schema for reliable downstream querying. • Built a queryable search interface in React over the standardized dataset, with resilient parsing, fallback selectors, and error handling validated through iterative testing on Linux/Unix development environments using Git version control.
Film Wizard Python, Pandas, NumPy, scikit-learn, Streamlit, TMDB Data
• Built a content-based recommendation pipeline in Python using Pandas and NumPy for data cleaning and feature preprocessing, with TF-IDF vectorization and cosine-similarity scoring across structured metadata. • Tested and validated recommendation outputs through an interactive Streamlit interface, iterating on preprocessing logic to improve result relevance.
ApplyIntel Next.js, TypeScript, FastAPI, Python, SQLAlchemy, PostgreSQL, Chart.js
• Designed and developed a full-stack data-management platform with CRUD workflows, search, filtering, sorting, pagination, CSV export, follow-up tracking, and analytics dashboards for structured job-application data. • Engineered REST APIs, relational models, validation logic, and analytics services using FastAPI, SQLAlchemy, and PostgreSQL-compatible storage; integrated the backend with a responsive Next.js interface using Axios and Chart.js. • Configured environment-based deployment and local development workflows using Git, GitHub, Uvicorn, Vercel, Render, and Docker Compose while applying API testing, debugging, health checks, and technical documentation.
CelluloidFM JavaScript, Vue.js, Three.js, Web Audio API, Axios, Webpack, Babel, Sass
• Developed a browser-based radio platform that ingests live SomaFM station and track data, supports search, filtering, sorting, favorites, playlist export, deep-linked navigation, and responsive media interfaces. • Integrated Three.js and the Web Audio API to process browser frequency data and render synchronized audio visualizations; maintained a Webpack and Babel build workflow with modular Vue.js components.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL, HTML5, CSS3, C/C++ (coursework), Java (coursework), Bash/Shell
Data & AI: Pandas, NumPy, scikit-learn, data cleaning, data standardization, data alignment, TF-IDF, cosine similarity, generative AI tools, AI API integration, semantic search concepts
Frontend: React, Next.js, React Native, Vue.js, D3.js, Webpack, Bootstrap basics, Tailwind CSS, responsive web development
Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, SQLAlchemy, Axios, API integration
Databases: PostgreSQL, SQLite, SQL, relational data modeling, CSV/JSON processing
Development Tools & Practices: Git, GitHub, VS Code, Postman, Unix/Linux command line, Docker (basic), unit & integration testing, API validation, debugging, code reviews, Agile/Scrum, technical documentation